

AP Physics C: Mechanics

Practice Exam — Performance Report

Exam: May 4, 2026 • Timed (45 min) • 35 Multiple Choice Questions

11/35

Actual Score

31%

Percentage

26/35

Potential Score*

74%

Potential %

42:30

Time Used

17

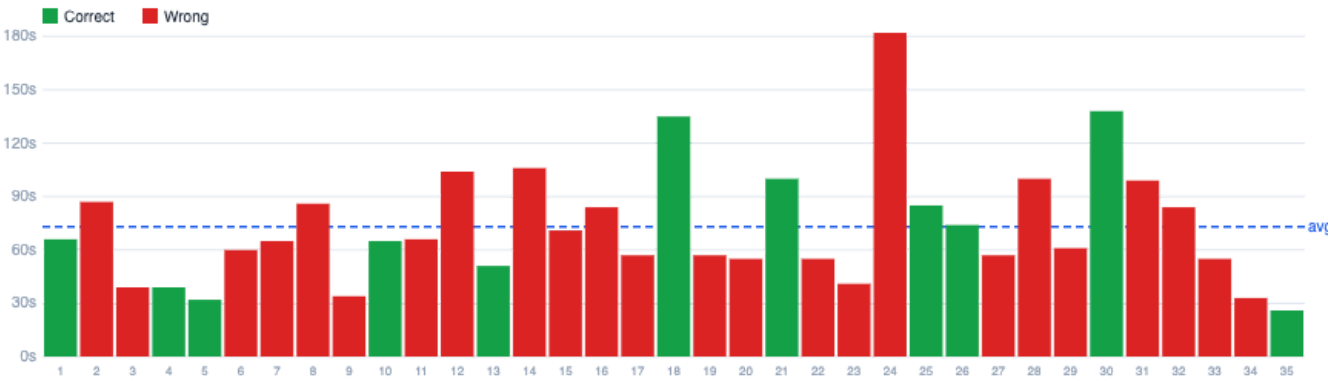
Flagged

*Potential = actual + 15 questions lost to minor mistakes (calculation, rush, sign errors — not knowledge gaps)

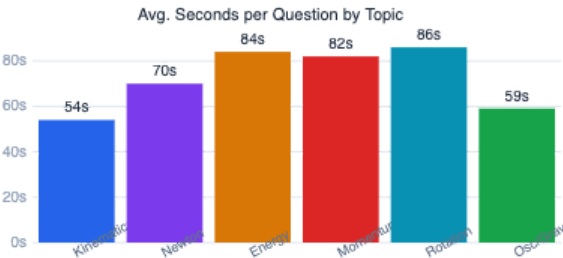
Topic Mastery

Kinematics	3/6 (50%)
Newton's Laws	1/6 (17%)
Work & Energy	2/6 (33%)
Momentum	1/6 (17%)
Rotation	3/6 (50%)
Oscillations & Grav.	1/5 (20%)

Time Per Question (seconds) — Correct vs Wrong



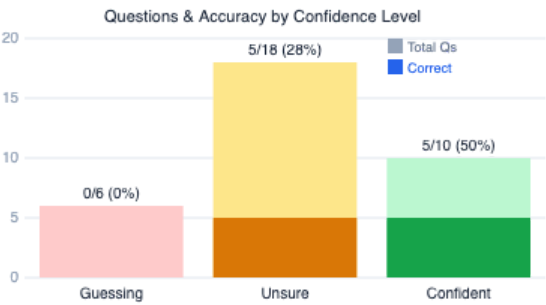
Average Time by Topic



Time Spent: Correct vs Wrong

	Correct (11)	Wrong (24)
Average	74s	72s
Median	66s	65s
Min	26s	33s
Max	138s	182s
Total	14m	29m

Confidence Calibration



Confidence Stats

0%

"Guessing" accuracy
6 questions, 0 correct

28%

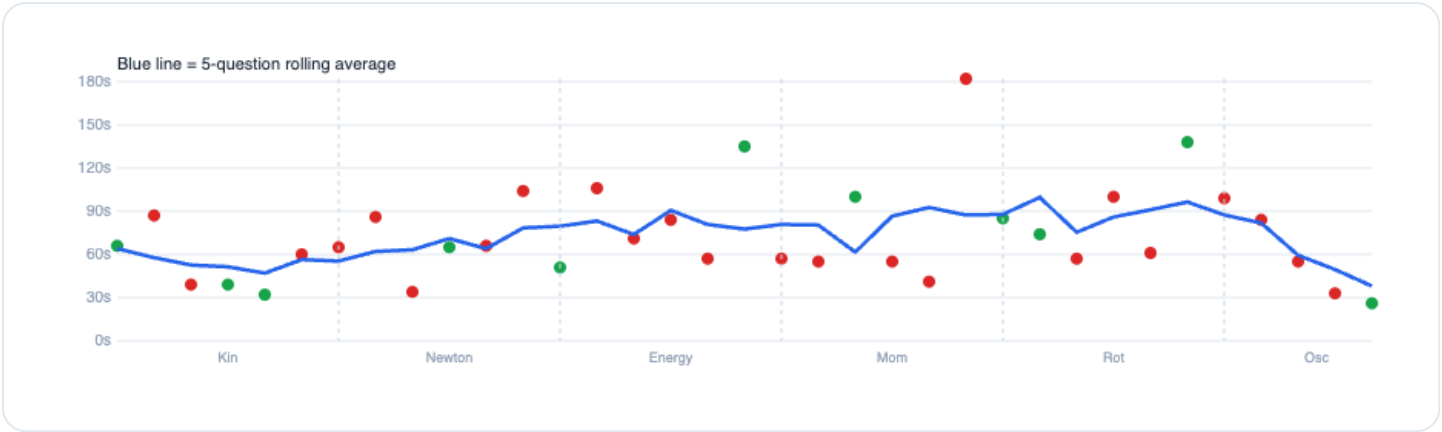
"Unsure" accuracy
18 questions, 5 correct

50%

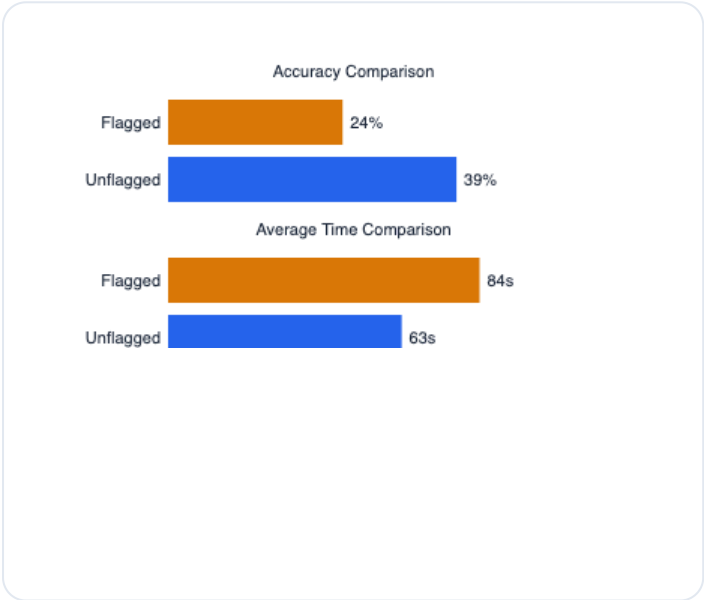
"Confident" accuracy
10 questions, 5 correct

5 questions marked "Confident" were answered incorrectly: Q6, Q9, Q11, Q15, Q17

Time Progression Through Exam



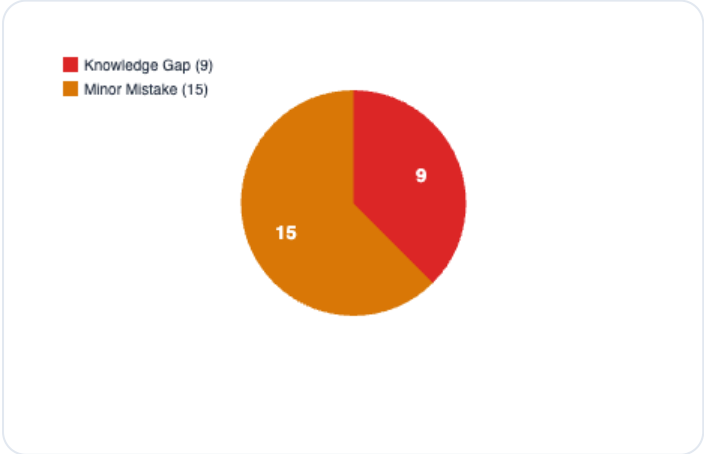
Flagged vs Unflagged Performance



Flagging Stats

Flagged questions	17
Unflagged questions	18
Flagged accuracy	4/17 (24%)
Unflagged accuracy	7/18 (39%)
Avg time (flagged)	84s
Avg time (unflagged)	63s

Error Classification



Breakdown (24 wrong answers)

Knowledge Gap	9 (37%)
Minor Mistake	15 (63%)

Minor mistakes include: calculation errors (5), sign/direction errors (4), formula application errors (3), rush/careless (2), vector addition error (1).
Fixing these yields the potential score of 26/35.

Knowledge Gaps (9 questions) — Concepts to learn

Q	TOPIC	GAP
Q11	Newton's Laws	Conical pendulum formula (confused with simple pendulum)
Q15	Work & Energy	Equilibrium stability — 2nd derivative test for $U(x)$
Q16	Work & Energy	Spring PE formula — placed the $1/2$ incorrectly
Q22	Momentum	Rocket equation — used linear approx instead of logarithmic
Q23	Momentum	Center of mass of solid hemisphere = $3R/8$
Q27	Rotation	Mass distribution affects rolling speed (solid vs hollow)
Q28	Rotation	Angular momentum conservation does not imply energy conservation
Q33	Oscillations	Escape speed = orbital speed $\times \sqrt{2}$
Q34	Oscillations	Physical pendulum formula $T = 2\pi\sqrt{I/mgd}$

Minor Mistakes (15 questions) — Knew the approach, lost points

Q	TOPIC	ERROR	TYPE
Q2	Kinematics	Half speed $\neq$ half height (forgot v -squared in KE)	Application
Q3	Kinematics	Picked centripetal accel for constant- g question	Careless
Q6	Kinematics	Used $v=at$ for non-constant acceleration (needed integral)	Application
Q7	Newton	Didn't decompose horizontal force along incline	Application
Q8	Newton	Found friction correctly, forgot to use total system mass	Calculation
Q9	Newton	Sign error: $T=M(g-a)$ instead of $T=M(g+a)$	Sign error
Q12	Newton	Added friction instead of subtracting (wrong direction)	Sign error
Q14	Energy	Missing factor of 2 from energy equation	Calculation
Q17	Energy	Knew $P=Fv$ but forgot $F=mg$, matched wrong answer 600W	Rush
Q19	Momentum	Used inelastic formula for elastic collision	Application
Q20	Momentum	Added perpendicular momenta algebraically, not vectorially	Application
Q24	Momentum	Averaged speeds instead of weighting by mass	Calculation
Q29	Rotation	Arithmetic error in KE fraction	Calculation
Q31	Oscillations	Subtracted wrong energy term in SHM equation	Calculation

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Q32	Oscillations	Inverted gravity ratio (got $g'=2g$ instead of $g/2$)	Calculation

Score Projection

SCENARIO	MC SCORE	PERCENTAGE	EST. AP SCORE
Actual	11/35	31%	1-2
Fix minor mistakes only	26/35	74%	5
Fix minor + knowledge gaps	35/35	100%	5

Complete Question Log

#	TOPIC	STUDENT	CORRECT	RESULT	CONFIDENCE	TIME	FLAG	ERROR TYPE
1	Kinematics	A	A	Correct	Confident	66s		—
2	Kinematics	A	C	Wrong	Unsure	87s	F	Minor
3	Kinematics	E	C	Wrong	Unsure	39s	F	Minor
4	Kinematics	C	C	Correct	Confident	39s		—
5	Kinematics	C	C	Correct	Confident	32s		—
6	Kinematics	A	C	Wrong	Confident	60s		Minor
7	Newton's Laws	A	C	Wrong	Unsure	65s	F	Minor
8	Newton's Laws	C	D	Wrong	Unsure	86s	F	Minor
9	Newton's Laws	B	D	Wrong	Confident	34s		Minor
10	Newton's Laws	B	B	Correct	Confident	65s		—
11	Newton's Laws	A	B	Wrong	Confident	66s		Knowledge
12	Newton's Laws	A	B	Wrong	Unsure	104s	F	Minor
13	Work & Energy	B	B	Correct	Confident	51s		—
14	Work & Energy	A	C	Wrong	Guessing	106s	F	Minor
15	Work & Energy	C	A	Wrong	Confident	71s		Knowledge
16	Work & Energy	D	C	Wrong	Guessing	84s	F	Knowledge
17	Work & Energy	A	D	Wrong	Confident	57s		Minor
18	Work & Energy	B	B	Correct	Unsure	135s	F	—
19	Momentum	C	B	Wrong	Unsure	57s	F	Minor
20	Momentum	A	B	Wrong	Guessing	55s	F	Minor
21	Momentum	C	C	Correct	Unsure	100s	F	—
22	Momentum	A	B	Wrong	Guessing	55s	F	Knowledge
23	Momentum	A	C	Wrong	Unsure	41s	F	Knowledge
24	Momentum	D	C	Wrong	Unsure	182s	F	Minor

#	TOPIC	STUDENT	CORRECT	RESULT	CONFIDENCE	TIME	FLAG	ERROR TYPE
25	Rotation	C	C	Correct	Unsure	85s	F	—
26	Rotation	C	C	Correct	Unsure	74s	F	—
27	Rotation	C	A	Wrong	Unsure	57s	F	Knowledge
28	Rotation	D	B	Wrong	Unsure	100s		Knowledge
29	Rotation	A	C	Wrong	Guessing	61s		Minor
30	Rotation	A	A	Correct	Unsure	138s		—
31	Osc & Grav	E	B	Wrong	Guessing	99s		—
32	Osc & Grav	B	D	Wrong	Unsure	84s		Minor
33	Osc & Grav	B	C	Wrong	Unsure	55s		Knowledge
34	Osc & Grav	A	C	Wrong	Unsure	33s		Knowledge
35	Osc & Grav	B	B	Correct	—	26s		—

Additional Analytics

Answer Letter Distribution



Time vs Question Number (Fatigue Check)

